

## **Relationship between Chamber Advantage and Legislative Productivity**

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### **Executive Summary:**

This paper tackles and investigates the relationship between how Chamber advantage affects the amount of Legislative productivity that occurs in Congress. Chamber advantage refers to when one legislative chamber has a higher number of members in congress, while legislative productivity encompasses multiple factors that go into producing and finalizing legislatures. In order to do this we utilized official legislative data collected that dates from congressional meetings number eighty through one hundred and twelve. In the research we will share the findings that we came upon showcasing that there is a direct relationship between the chamber advantage and the amount of legislative productivity that occurs as a result.

### **Introduction:**

The United States Congress is a bicameral legislature that is composed of the House of Representatives and the Senate which both contribute to legislative governance. Each state elects representatives and senators that will contribute to deciding whether or not to enact laws, create policies, and address the needs of citizens. The chambers of congress have the power to investigate whether or not a policy or law will make a significant impact on citizens.

Members of congress spend a lot of time having to hold hearings or even conduct investigations in committees. When taking into account the House of Representatives there are 435 seats to which members are elected to serve two year terms and make decisions based on the size of state populations. The Senate holds 100 senators with two senators from each state that serve six year terms to make decisions. Both chambers play significant roles to which the house represents individual districts and the senate represents states. Among the various factors that both chambers play within congress, the chamber advantage has become a notable factor that can contribute to structural differences that shape the dynamic of lawmaking within each chamber.

The chamber advantage can be seen as a significant element that influences the legislative process. There are contrasting rules and procedures in which each chamber has to follow which may contribute to the differences in chamber advantages along with the ways in which laws and policies are enacted. In this sense, the aspect of productivity is significant when addressing the question of How does the size of a party's seat share advantage in a chamber affect the legislative productivity of that chamber? This paper seeks to investigate the relationship between how Chamber advantage affects the amount of Legislative productivity that occurs in Congress. In order to do so, official legislative data will be used to see dates from congressional meetings number eighty through one hundred and twelve then compare it to see if there is a relationship between the chamber advantage and legislative productivity.

### **Data Description:**

In this study we used two files: congress.csv & legislation.csv to experiment and look for effects between having the chamber advantage and changes in legislation productivity. Both of the files each had data regarding congressional meetings eighty through one hundred and twelve, however neither had enough data to analyze whether there was a relationship. As such in order for us to observe the effect we merged the two files together omitting for congress meeting 113 as the data set had missing variables at the time.

From the dataset congress.csv we were able to obtain information on the amount of seats each party held and whether or not they had control on house, senate, and presidency. For the purpose of our study we mainly focused on the information that they gave about the senate. The dataset originally had the variable scontrol that either gave D for Democrat controlled or R for Republican controlled. In order for us to look for whether there was an effect on this we opted to turn it into a new variable that we created, chamberadv, in which 0= Republican and 1=Democratic. We also made another variable, dwnom\_mean, in which if the value is positive there is a more Republican average ideology and Democratic if it is negative. The other file that we used was legislation.csv which gave us the amount of landmark legislation that was passed for each corresponding congress meeting.

In order to create the linear regression that allowed us to look at the effect we used our three key variables: our new variable chamberadv which tells us whether there is a chamber advantage and the n\_landmark that tells us how much legislation bills were passed and later on dwnom\_mean to calculate the average political ideology for each congressional meeting.

| cong             | year            | dsh               | rsh              | osh              | dss           |
|------------------|-----------------|-------------------|------------------|------------------|---------------|
| Min. : 80        | Min. :1946      | Min. :189.0       | Min. :140.0      | Min. :0.0000     | Min. :45.00   |
| 1st Qu.: 88      | 1st Qu.:1962    | 1st Qu.:212.0     | 1st Qu.:174.0    | 1st Qu.:0.0000   | 1st Qu.:48.00 |
| Median : 96      | Median :1978    | Median :249.0     | Median :186.0    | Median :0.0000   | Median :54.00 |
| Mean : 96        | Mean :1978      | Mean :243.9       | Mean :190.8      | Mean :0.3939     | Mean :53.42   |
| 3rd Qu.:104      | 3rd Qu.:1994    | 3rd Qu.:263.0     | 3rd Qu.:221.0    | 3rd Qu.:1.0000   | 3rd Qu.:57.00 |
| Max. :112        | Max. :2010      | Max. :295.0       | Max. :245.0      | Max. :2.0000     | Max. :68.00   |
| rss              | oss             | scontrol          | hcontrol         | pcontrol         |               |
| Min. :32.00      | Min. :0.0000    | Length:33         | Length:33        | Length:33        |               |
| 1st Qu.:41.00    | 1st Qu.:0.0000  | Class :character  | Class :character | Class :character |               |
| Median :45.00    | Median :0.0000  | Mode :character   | Mode :character  | Mode :character  |               |
| Mean :45.15      | Mean :0.5758    |                   |                  |                  |               |
| 3rd Qu.:51.00    | 3rd Qu.:1.0000  |                   |                  |                  |               |
| Max. :55.00      | Max. :2.0000    |                   |                  |                  |               |
| d_mean           | r_mean          | d_median          | r_median         | d_sd             |               |
| Min. : -0.3841   | Min. : 0.1971   | Min. : -0.3930    | Min. : 0.1795    | Min. : 0.1243    |               |
| 1st Qu.: -0.3348 | 1st Qu.: 0.2258 | 1st Qu.: -0.3410  | 1st Qu.: 0.2140  | 1st Qu.: 0.1495  |               |
| Median : -0.2900 | Median : 0.2504 | Median : -0.3085  | Median : 0.2440  | Median : 0.1871  |               |
| Mean : -0.2856   | Mean : 0.3428   | Mean : -0.2993    | Mean : 0.3347    | Mean : 0.1754    |               |
| 3rd Qu.: -0.2516 | 3rd Qu.: 0.4742 | 3rd Qu.: -0.2880  | 3rd Qu.: 0.4635  | 3rd Qu.: 0.1958  |               |
| Max. : -0.1463   | Max. : 0.7072   | Max. : -0.1230    | Max. : 0.7230    | Max. : 0.2120    |               |
| r_sd             | n_landmark      | dwnom_mean        | chamberadv       |                  |               |
| Min. : 0.1399    | Min. : 7.00     | Min. : -0.10243   | Min. : 0.0000    |                  |               |
| 1st Qu.: 0.1542  | 1st Qu.:15.00   | 1st Qu.: -0.03638 | 1st Qu.: 0.0000  |                  |               |
| Median : 0.1609  | Median :19.00   | Median : 0.02372  | Median :1.0000   |                  |               |
| Mean : 0.1601    | Mean :18.88     | Mean : 0.05724    | Mean : 0.6667    |                  |               |
| 3rd Qu.: 0.1689  | 3rd Qu.:22.00   | 3rd Qu.: 0.13184  | 3rd Qu.:1.0000   |                  |               |
| Max. : 0.1745    | Max. :31.00     | Max. : 0.34552    | Max. :1.0000     |                  |               |

Figure 1: Summary statistics for each of values in the combined dataset (join).

### Description of Statistical Methods:

While trying to investigate the relationship between chamber advantage and legislative productivity, one of the statistical methods we used in our research was a simple linear regression analysis. The purpose of linear regression is to be able to model the relationship between a dependent variable and one or more independent variables which can be calculated through the formula  $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ .  $Y_i$  means the estimated or predicted Y value for observation i.

This can also be known as the dependent or response variable and in our case our dependent variable is “n\_landmark” to represent the number of legislation passed.  $\beta_0$  is an intercept that

represents the estimate of regression.  $\beta_1$  is the estimate of the regression slope and  $X_i$  is the value of  $X$  for observation  $i$  which can also be known as the independent or predictor variable and is represented by “chamberadv” in our study to show chamber advantage with  $\varepsilon_i$  being the random error term.

In addition to this method for our investigation, we also used a form of advanced analysis to help us better understand the predictive power and significance of the relationship between chamber advantage “chambadv” and legislative productivity “n\_landmark.” The method used was a multiple regression with the added variable dwnom\_mean making it a fixed effect. This method showed that as of the moment there is no major statistical difference between chamber advantage and legislative productivity, however, there is a possibility that if we had accounted for more controlled variables we could have found a positive relationship. As of right now there is none.

### Results:

From our linear regression model (shown in figure 2) it tells us that a one unit increase in chamber advantage is associated with a 2 point increase in legislative productivity, however our p-value states that it is statistically insignificant and not completely accurate.

```

Residuals:
    Min       1Q   Median       3Q      Max
-12.5455  -4.5455   0.4545   3.4545  11.4545

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   17.545      1.816   9.663 7.22e-11 ***
chamberadv     2.000      2.224   0.899   0.375
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.022 on 31 degrees of freedom
Multiple R-squared:  0.02543,    Adjusted R-squared:  -0.006008
F-statistic: 0.8089 on 1 and 31 DF,  p-value: 0.3754

```

**Figure 2: Regression results of the effect between chamber advantage & legislative productivity**

As such we are unable to statistically say that there is a strong relationship between chamber advantage and legislative productivity from our datasets.

Due to this we wanted to look if controlling for political ideology would change any results. As such we ran a multiple linear regression (shown in figure 3) that tells us if we control for political ideology a one unit increase in chamber advantage is associated with .42 point decrease in legislative productivity, however our p-values still states that this is statistically insignificant.

```

Residuals:
    Min       1Q   Median       3Q      Max
-12.7810  -2.7588   0.1704   3.0305   9.0768

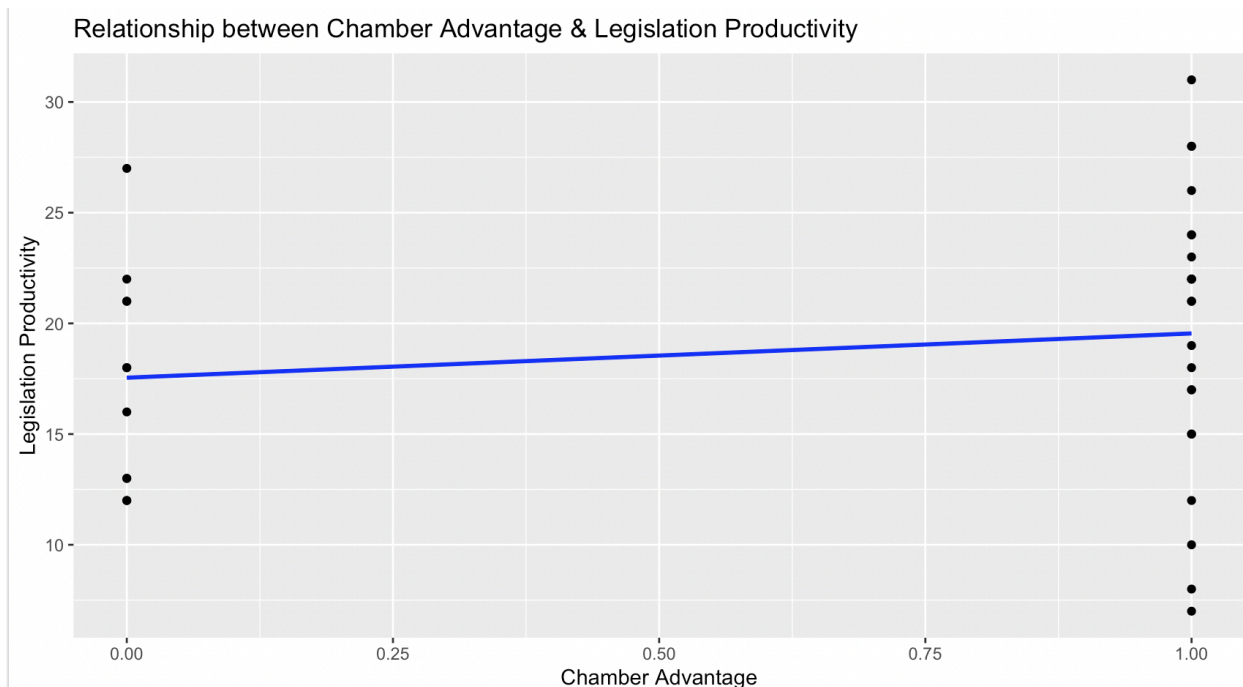
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  20.7792     1.7271  12.031 5.23e-13 ***
chamberadv   -0.4199     1.9525  -0.215  0.83117
dwnom_mean  -28.3090     7.3461  -3.854  0.00057 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.007 on 30 degrees of freedom
Multiple R-squared:  0.3481,    Adjusted R-squared:  0.3047
F-statistic:  8.01 on 2 and 30 DF,  p-value: 0.001631

```

**Figure 3: Regression of chamber advantage & legislative productivity controlling for political ideology**

Additionally another thing we did was creating a scatterplot in order to showcase the relationship between the two variables adding in a line of best fit shown in figure 4. From this we can see that there are usually more times where the senate is more likely to have a Democratic advantage, but we cannot say that there is a higher legislative productivity because our p-value doesn't cross the threshold.



**Figure 4: Linear regression plot of Chamber advantage & Legislative productivity**

### **Conclusion:**

Ultimately, the United States Congress is a complex system that has allowed for the House of Representatives and the Senate to shape the ways in which policies and laws are enacted. This research aimed to address the relationship between how Chamber advantage affects the amount of Legislative productivity that occurs in Congress. When looking at the differences in the House of Representatives and the Senate, the chamber advantage was presented as a possible influential factor in legislative productivity. However, the aspect of the chamber advantage did not necessarily translate into productivity.

The differences in party seat advantages present the idea that a chamber has influence on productivity yet the results from the data show inconclusive results. The data sets that were used to analyze the effect of chamber advantage and legislative productivity showed constraints when addressing the association between the variables. There may be a positive relationship between chamber advantage and legislative productivity, but based on the data there are no controlling variables that could demonstrate whether or not there is a direct impact. In future research it would be significant to seek control variables that would provide more accurate results that could support the hypothesis of chamber advantage influencing legislative productivity.



R code: All

CODE:

```
# Set the working directory
```

```
setwd("/Users/matthew/Downloads/UCSD/Poli 100X/Research")
```

```
rm(list=ls())
```

```
# Load the csv data file & packages
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
congress <- read.csv("congress.csv")
```

```
legislation <- read.csv("legislation.csv")
```

```
congress2 <- na.omit(congress)
```

```
#merge
```

```
join <- full_join(congress2, legislation, by = "cong")
```

```
# create a variable to look for chamber advantage based off senate controlling party
```

```
join$chamberadv <- ifelse(join$scontrol == "R", 0, ifelse(join$scontrol == "D", 1, NA))
```

```
# looks for effect on legislative productivity and chamber advantage
```

```
leg_product <- lm(chamberadv ~ n_landmark, data= join)
```

```
# summarize the linear regression
```

```
summary(leg_product)
```

```
#plot
```

```
joinlm <- ggplot(join, aes(x= chamberadv, y= n_landmark)) + geom_point()
```

```
+ xlab("Chamber Advantage") + ylab("Legislation Productivity")
```

```
+ geom_smooth(method="lm", se=FALSE, linetype="solid", color="blue")
```

```
#print plot
```

```
print(joinlm+ ggtitle("Relationship between Chamber Advantage & Legislation Productivity"))
```